

Tracking Cognition through Emotion: Designing a Device-Light Emotion Selection Routine for Daily Cognitive Engagement in Older Adults

Xiaoyu Ren
Tianjin University
Tianjin, China
renxiaoyu1919@163.com

Yibo Zhang[†]
Cheongju University
Chungbuk, Republic of Korea
zhangyibo1919@outlook.com

Abstract

Cognitive decline in older adults often progresses subtly, making it difficult to detect in everyday life. To support aging-in-place, we present Feeling Tracker, a lightweight, device-light system that promotes daily cognitive engagement through emotion logging. Users select an emoji representing their mood, followed by an optional short reflection via text or voice. Visualizations in a calendar interface help surface emotional patterns and episodic memory cues. Unlike commercial mood-tracking tools, Feeling Tracker embeds frictionless interaction into daily routines using fixed-location devices, requiring minimal effort or technical skill. A three-week deployment with five independently living older adults showed high adherence and reported emotional and cognitive benefits. Participants described the activity as both intuitive and meaningful, reinforcing memory through low-effort self-reflection. Our design contributes a novel cognitive proxy mechanism for aging populations, shifting mood tracking from general self-monitoring to routine-based cognitive support. The system may also be adaptable for users with mild cognitive impairment (MCI), offering future avenues for inclusive, accessible digital health interventions.

CCS Concepts

· Human-centered computing → Ubiquitous and mobile computing systems and tools; · Human-centered computing → Empirical studies in ubiquitous and mobile computing; · Applied computing → Health care information systems.

Keywords

Cognitive engagement, emotion tracking, low-friction interaction, ubiquitous computing, device-light design, memory support

[†] Contact author

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from Permissions@acm.org.
UbiComp Companion '25, Espoo, Finland.

© 2025 Copyright is held by the owner/author(s). Publication rights licensed to ACM.

ACM ISBN 979-8-4007-1481-8/25/10.

<https://doi.org/10.1145/3714394.3754360>

ACM Reference format:

Xiaoyu Ren and Yibo Zhang. 2025. Tracking Cognition through Emotion: Designing a Device-Light Emotion Selection Routine for Daily Cognitive Engagement in Older Adults. In *Companion of the 2025 ACM International Joint Conference on Pervasive and Ubiquitous Computing (UbiComp Companion '25)*, October 12–16, 2025, Espoo, Finland. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3714394.3754360>.

1 Introduction

Maintaining cognitive health is a key challenge in an aging society [1]. Regular mental stimulation and reflection are known to help older adults preserve memory and engagement [2]. In particular, daily journaling or emotional self-report can provide older people an opportunity to reflect on their experiences, contributing to sustained cognitive function and social connectivity [3]. Giving seniors an active role in monitoring their emotional well-being is also empowering - self-reported reflection offers therapeutic benefits and a sense of control [4]. However, traditional cognitive exercises or technology-driven interventions (e.g., wearable cognitive monitors) can feel intrusive or burdensome to older adults [5]. There is a need for unobtrusive, easily-adoptable methods to support memory and cognition in everyday routines.

Many older adults prefer to age in place with minimal disruption, valuing solutions that respect their independence and privacy [6]. Yet few interfaces have been tailored for logging emotions or memories in home settings for seniors. Wearable devices and high-tech sensors often face adoption barriers among older users - concerns about complexity, comfort, cost, and stigma can deter sustained use [7]. This motivates a “device-light” approach that avoids wearables altogether, instead leveraging simple self-report measures. By using familiar emotive cues (emojis) and lightweight input, we aim to engage older adults in daily cognitive reflection without specialized hardware. Prior work suggests that asking individuals to recall their feelings on a daily basis can even serve as a memory aid, reducing forgetfulness over longer periods [8]. Building on these insights, we designed Feeling Tracker (Shown in Figure 1), a lightweight “emotion journal” accessible via portable digital devices (e.g., tablets or smartphones), supporting daily self-reflection in home settings for older adults. Our goal is to spark recollection and mild cognitive exercise through the simple act

of emotion selection and optional reminiscence. We describe the design of the Feeling Tracker system and report findings from a preliminary deployment, illustrating how daily emotion logging can act as a proxy for memory activation. The work contributes a design-led perspective on supporting cognitive engagement in older adults via non-invasive, everyday technology.

Feeling Tracker Aging Mood Support System

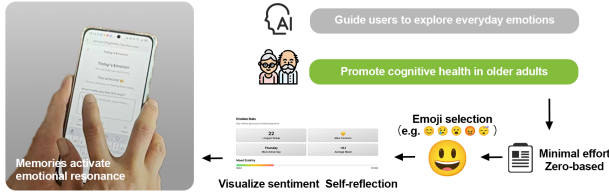


Figure 1: Overview of the Feeling Tracker Aging Mood Support System.

2 The Feeling Tracker System

2.1 Daily Emotion Selection and Reflection

The Feeling Tracker is a simple emotion journaling system that users can incorporate into their everyday routine (Shown in Figure 2). Each day, the user is prompted to “record how they feel” using an emoji-based selection. The interface provides a grid-based selection of familiar emoticon icons representing predefined emotional states (e.g., Happy, Sad, Neutral, etc.) in a grid for easy selection. This emoji-driven input requires no typing or complex interaction, reducing cognitive and technical barriers. After choosing an emoji that best matches their current mood, the user is invited to “share a short memory or context” for that feeling. They can write a brief reflection about what made them feel that way, though this step is optional to keep the friction low. For example, a user might select a smiling face for “Happy” and note, “Had a wonderful walk in the park and chatted with my grandchildren.” The interface emphasizes rapid, intuitive input through single-tap emotion selection and streamlined text entry for capturing user reflections. This design encourages “daily self-reflection” while respecting the user’s effort limits. Even if the user opts not to type a story, the act of internally recalling why they feel a certain way provides a moment of cognitive engagement. Emoji selection was grounded in Unicode’s “Most Recognized” set; pilot validation ($n = 5$) yielded 98% recognition accuracy and an average selection time of 2.3s ($SD = 0.7$), significantly faster than text entry (8.9s, $SD = 2.1$), confirming their efficacy as low-cognitive-load affect proxies despite inherent expressiveness constraints. Unlike typical mood apps, Feeling Tracker focuses on cognitive stimulation for older adults through routine-based, low-effort reflection—making it distinct from commercial tools designed for general self-tracking.

2.2 Reviewing Emotions Over Time

All daily entries are stored in a history view that allows users to revisit past emotional states and associated reflections. The history can be accessed in two formats: a chronological list view,

where each entry displays the date, selected emoji, and optional memory note (e.g., “May 20, 2025 - Neutral - Just an ordinary day, watched TV and read a book”), and a calendar view, where emojis populate each day’s cell, offering a visual summary of emotional patterns over the month. These lightweight visualizations support both self-awareness and reminiscence. By browsing past days, users may recall forgotten moments or observe trends in their emotional rhythms—such as stretches of positive moods or alternating states. Several participants described this experience as “looking back at how the month felt,” indicating its value for emotional insight and memory reinforcement. The interface design prioritizes clarity, using large icons, high-contrast text, and a clean layout to accommodate visual or cognitive limitations common in older adults. Complex menus and layered navigation were deliberately avoided to preserve ease of use. Importantly, the system is intended to operate on everyday home devices such as tablets or smartphones placed in consistent locations, but it may also be adapted to ambient or shared digital formats (e.g., smart displays, digital calendars) to suit varying levels of digital comfort. By integrating seamlessly into daily routines and requiring only a minute of attention, the history feature contributes to a low-effort, sustainable engagement loop—reinforcing the system’s role as a subtle yet meaningful cognitive and emotional support tool for aging-in-place.

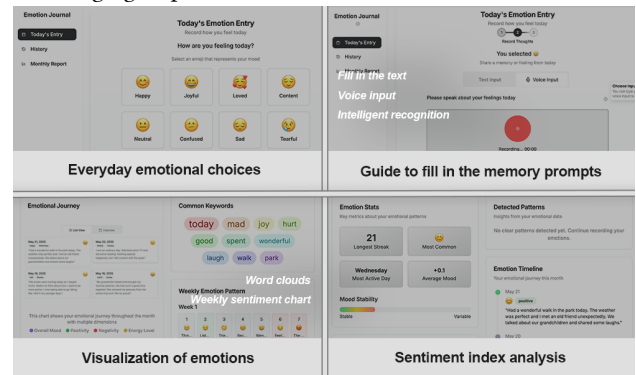


Figure 2: Overview of the Feeling Tracker system interface, illustrating four core modules: (1) daily emotion selection, (2) guided memory reflection with optional voice input, (3) visualization of emotional trends via word clouds and sentiment charts, and (4) sentiment index analysis for longitudinal engagement insights.

2.3 Architectural Innovation

The core innovation of Feeling Tracker lies in its device-light architecture, characterized by three design principles: 1. Zero Physical Burden Devices remain fixed in habitual locations (e.g., bedside tables), eliminating wearable discomfort and charging requirements reported in prior studies. 2. Cognitive Proxy Mechanism Daily emotion selection serves as a deliberate trigger for episodic memory retrieval. Neuroscience evidence confirms emotional valence enhances autobiographical recall [9]. 3. Frictionless Interaction Protocol [10] Strict adherence to ≤ 1 s

primary interaction (emoji selection) and ≤ 30 s optional reflection aligns with older adults' attentional capacity. This paradigm shifts ubiquitous computing from "always-on" wearables to "ambiently available" environmental anchors - a critical advance for aging populations.

3 Pilot Study

We conducted a pilot deployment of the Feeling Tracker with a small group of older adults to understand usage patterns and gather feedback about their experiences. The study examined whether participants would adopt the daily emotion selection routine and how it affected their engagement and reflections.

3.1 Deployment and Usage Data

Participants: Five older adults (4 female, 1 male), aged 68-75, living independently in the community were recruited. None had prior experience with mood tracking apps, and their comfort with smartphones or tablets ranged from low to moderate. We provided access to the Feeling Tracker via a simplified interface installed on either a tablet or smartphone, depending on the participant's preference. The device was placed in a familiar, easily accessible location at home (e.g., kitchen table, bedside stand). Participants were instructed to log their daily emotion at a consistent time of their choosing (e.g., after dinner or before bedtime) for a period of three weeks (Shown in Figure 3).

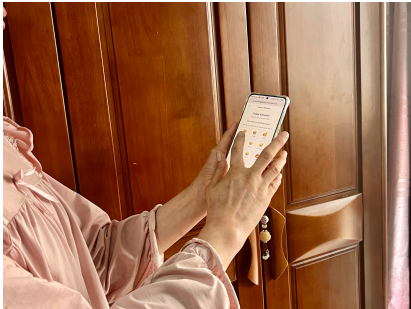


Figure 3: An older adult interacting with the Feeling Tracker app on a smartphone as part of a daily emotion journaling routine.

Usage Results: Overall adherence to the routine was high. Participants completed 90% of the intended daily entries on average, with three participants logging an entry every single day of the 21-day trial. The remaining two missed a few days due to travel or forgetting, but resumed afterward. Over the pilot, a total of 98 emotion entries were collected. Most participants used a range of different emojis over the weeks, rather than selecting the same emotion every time. This suggests they were paying attention to how they truly felt each day and not just defaulting to a single response. On average, participants chose positive emotions ("Happy", "Content", "Joyful") 53% of the days, neutral or mixed feelings 28%, and negative emotions ("Sad", "Angry", "Exhausted") 19% of the days. Every participant wrote at least a brief text reflection for the majority of their entries (about 70% of the days they logged). These reflections

were usually one or two sentences long, often describing a specific event (e.g., a visit from family, a difficult night's sleep) that influenced their mood. The act of writing did not significantly deter usage—entries without text were mostly on days rated "Neutral" or when participants were especially tired. Importantly, no one reported confusion about how to use the interface after the first day; this indicates that the low-friction design was indeed intuitive for users with diverse tech backgrounds.

3.2 Participant Insights

After the trial, we interviewed participants to understand their perceptions of the routine. Overall, feedback was very positive. All five participants expressed that they found the daily emotion log easy to use and enjoyable. One participant (P3, age 72) shared, "It became part of my day. Each night I would think about what happened and how I felt. It was nice to have that moment to reflect." This sentiment illustrates how the routine served as a gentle nudge for self-reflection and memory recall, effectively becoming a cognitive exercise. Participants reported that writing a short memory was the most meaningful part for them. P5 (age 75) said she looked forward to writing "a little story each day", and that it gave her conversations to share with her daughter later. Some also noted that on more challenging days, articulating a feeling (even just selecting "Sad" or "Angry") was cathartic. For example, P2 recounted a day she felt lonely: "I chose the sad face. Seeing it there made me acknowledge it, and then I decided to call a friend. It pushed me to do something about it." Such anecdotes suggest the tool not only tracks mood but can prompt coping actions.

Participants appreciated the simplicity: "It's just one question - how do I feel - so it doesn't wear me out," said P4 (age 68), highlighting the low effort required. A few constructive comments emerged as well. Two participants mentioned they sometimes found it hard to pick one emoji if they had mixed emotions; they suggested maybe allowing multiple selections or an emoji that represents "mixed feelings." Another noted that if she skipped a day, the system didn't remind her (by design, we chose not to use disruptive notifications), but she thought an optional gentle reminder could be helpful. Despite these suggestions, all participants agreed they would like to continue using the Feeling Tracker beyond the study. In fact, at the end of the pilot, four out of five requested to keep the tablet or smartphone for personal use, indicating a strong acceptance of the routine. These findings gave us confidence that the approach is both feasible and valued by older users, even in a short deployment.

4 Discussion

Our pilot results demonstrate that the Feeling Tracker's approach resonates with older adults. We now reflect on the design features that contributed to this success, in terms of key ubiquitous computing values for aging-in-place: maintaining a low-friction experience, enabling device-light interaction, and embedding the routine into daily life.

4.1 Low-Friction Interaction

A primary design goal of the Feeling Tracker system was to minimize interactional friction for older adults. Through one-tap emoji selection and optional reflection input, the interface promotes daily engagement without cognitive burden. Notably, the system already supports voice input, enabling users to dictate their reflections instead of typing—an affordance especially appreciated by those with limited dexterity or visual acuity. Participants described this as “more natural than typing,” underscoring the importance of multi-modal input in inclusive design. By leveraging familiar metaphors (emojis), reducing navigational complexity, and integrating lightweight interaction within everyday routines, the system aligns with ubiquitous computing values: it becomes an unobtrusive part of home life, requiring minimal effort while encouraging sustained cognitive and emotional reflection.

4.2 Device-Light Design

The Feeling Tracker system embraces a device-light design philosophy: while it operates on a tablet or smartphone, the device itself is intended to remain fixed in a convenient and familiar location within the home—such as a bedside table, kitchen counter, or hallway shelf—rather than being carried or actively managed. This approach reflects older adults’ technology usage patterns, where many do not use smartphones regularly or feel confident with personal gadgets. By embedding interaction into a stationary device with a single-purpose interface, the system reduces cognitive and operational burden, making the experience more like using a daily calendar or analog journal.

Participants in our pilot often treated the system as part of their domestic routine. One user compared it to “checking the calendar on the fridge”, highlighting the familiarity and unobtrusiveness of its use. This integration exemplifies calm technology principles, where computation is present but not intrusive, and contributes to lowering the stigma sometimes associated with assistive or medical technologies. Future implementations could further reduce friction by incorporating the system into ambient smart displays or voice-activated routines, allowing for even more passive yet engaging interaction. Ultimately, the goal is not to eliminate devices entirely, but to ensure that technology supports rather than disrupts the rhythms of aging in place.

4.3 Embedded Routine for Aging-in-Place

Finally, the Feeling Tracker demonstrates the power of embedding a health intervention into an existing daily routine. Rather than asking older adults to set aside time for “brain training” or to attend a class, we piggyback on the natural daily cycle – specifically, the common habit of reflecting on one’s day in the evening or noting the date in the morning. By tying mood journaling to these regular activities (for instance, right after dinner cleanup, as one participant did), the routine becomes sustainable. Research on medication adherence echoes this

strategy: older adults are most successful in taking medications when they incorporate them into routines like meals or bedtime [11]. In our pilot, participants quickly found a groove for using the tracker at a consistent time, and their high compliance (90% daily logging) reflects the effectiveness of routine embedding. An embedded approach supports aging-in-place by fitting into the rhythms of home life rather than disrupting them. It leverages context – e.g. seeing the device on the kitchen counter is a passive reminder to log mood after breakfast. Over time, this could form a habit that persists without external enforcement. Embedding also means the activity becomes personalized and meaningful; as participants reported, “it became part of my day,” suggesting it was internalized as more than a study task. For aging-in-place technologies, this is a critical insight: interventions should become a seamless part of one’s lifestyle, not an additional chore. In the long run, a daily emotion routine could potentially provide early warnings for cognitive or emotional changes (e.g., noticing increasing days of confusion or apathy), all while the individual remains comfortably at home. By building on the user’s own routine and environment, the Feeling Tracker fosters autonomy and consistent engagement, which are key to supporting independent living in older age.

4.4 Extension to Mild Cognitive Impairment

While our pilot participants were cognitively healthy, the simplicity and routine-based design of Feeling Tracker suggests potential for use by individuals with mild cognitive impairment (MCI). Future work should explore its usability in this group, considering visual clarity, simplified emoji sets, and possible caregiver-assisted logging.

Conclusion

We presented the Feeling Tracker System, a novel device-light daily emotion selection routine to engage older adults in reflective cognitive activity. Our design and pilot study illustrate how a simple, low-effort interaction can be woven into everyday life to support aging in place. Participants embraced the routine, suggesting that even minimalist interventions can have a meaningful impact on their daily cognitive engagement and emotional awareness. Moving forward, we plan to explore long-term use of the system in home settings and investigate whether patterns in emotion entries correlate with cognitive health markers. This work contributes to the ubiquitous computing and HCI literature by demonstrating the value of emotion-driven, routine-based design for older adult well-being – technology that does not feel like technology, fostering health and cognition one day, and one emotion, at a time.

References

- [1] C. Chen, S. Ding, and J. Wang. 2023. Digital health for aging populations. *Nature Medicine* 29, 7 (2023), 1623–1630. <https://doi.org/10.1038/s41591-023-02391-8>
- [2] C. J. Dinius, C. E. Pocknell, M. P. Caffrey, and R. A. P. Roche. 2023. Cognitive interventions for memory and psychological well-being in aging and dementia. *Frontiers in Psychology* 14 (2023), Article 1070012. <https://doi.org/10.3389/fpsyg.2023.1070012>
- [3] R. Maharjan, K. Doherty, D. A. Rohani, P. Bækgaard, and J. E. Bardram. 2022. Experiences of a Speech-enabled Conversational Agent for the Self-report of

- Well-being among People Living with Affective Disorders: An In-the-Wild Study. *ACM Transactions on Interactive Intelligent Systems* 12, 2, Article 10 (June 2022), 29 pages. <https://doi.org/10.1145/3484508>
- [4] M. Sykora, S. Elayan, L. Angelini, C. Röcke, M. El Kamali, E. Mugellini, and S. Guye. 2021. Understanding older adults' affect states in daily life for promoting self-reflection about mental wellbeing. In G. Andreoni and C. Mambretti (Eds.), *Digital Health Technology for Better Aging: A Multidisciplinary Approach*. Springer, Cham, 179–193. https://doi.org/10.1007/978-3-030-72663-8_11
- [5] Z. Yumak and P. Pu. 2013. Survey of Sensor-Based Personal Wellness Management Systems. *BioNanoScience* 3, 3 (2013), 254–269. <https://doi.org/10.1007/s12668-013-0099-0>
- [6] D. Stones and J. Gullifer. 2016. 'At home it's just so much easier to be yourself': Older adults' perceptions of ageing in place. *Ageing & Society* 36, 3 (2016), 449–481. <https://doi.org/10.1017/S0144686X14001214>
- [7] C. Wu and G. G. Lim. 2024. Investigating older adults users' willingness to adopt wearable devices by integrating the technology acceptance model (UTAUT2) and the technology readiness index theory. *Frontiers in Public Health* 12 (2024), Article 1449594. <https://doi.org/10.3389/fpubh.2024.1449594>
- [8] J. Fogler and L. Stern. 2014. *Improving Your Memory: How to Remember What You're Starting to Forget*. Johns Hopkins University Press, Baltimore, MD.
- [9] Wardell V, Palombo D J. 2024. Stability and malleability of emotional autobiographical memories. *Nature Reviews Psychology*, 3, 6 (2024): 393-406. <https://doi.org/10.1038/s44159-024-00312-1>
- [10] Kemper J, Jankowski S. 2024. Silicon Valley's Frictionless Future: The Design Philosophy of Frictionlessness. *The De Gruyter Handbook of Automated Futures: Imaginaries, Interactions and Impact*, 2(2024): 287. <https://doi.org/10.1515/9783110792256-018>
- [11] M. J. Sanders and T. Van Oss. 2013. Using daily routines to promote medication adherence in older adults. *The American Journal of Occupational Therapy* 67, 1 (2013), 91–99. <https://doi.org/10.5014/ajot.2013.005033>